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IOT and Applications (3160716)

Summer – 2025 – Solution

Question : 01

Q.1 (a) Describe Internet of Things.

(3 Marks)

Internet of Things (IoT)

- The **Internet of Things (IoT)** is a network of physical devices that are connected to the internet and can collect, exchange, and process data automatically without human intervention.
- These devices contain:
 - Sensors
 - Software
 - Communication modules
 - Actuators
- IoT enables devices to communicate with each other and perform smart operations.

Working of IoT

1. Sensors collect data from the environment.
2. Data is sent through the internet.
3. Cloud/server processes the data.
4. Device takes action using actuators.

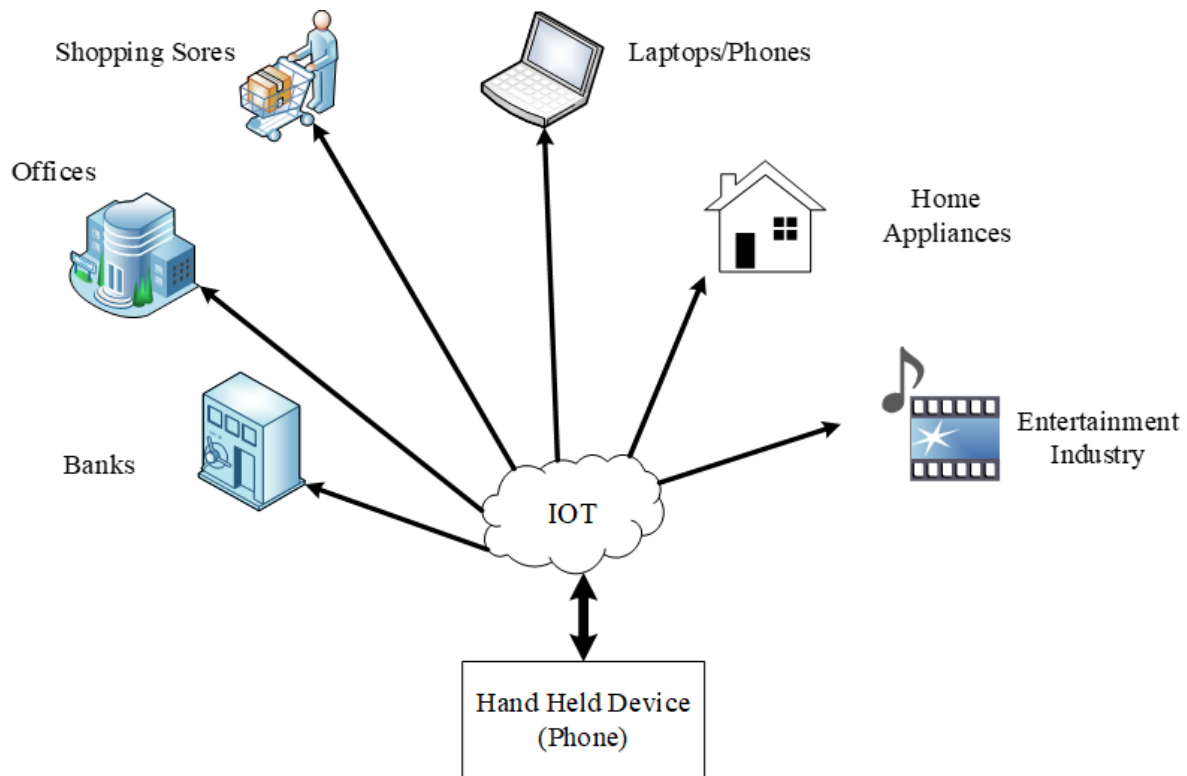
Applications of IoT

- Smart Home
- Smart City
- Healthcare Monitoring
- Industrial Automation
- Smart Agriculture

Advantages of IoT

- Automation
- Real-time monitoring
- Improved efficiency
- Reduced human effort

Simple IoT Architecture Diagram



Conclusion

IoT connects smart devices through the internet to make systems intelligent, automated, and efficient.

Q.1 (b)

(i) List the sources of IoT development board which can be used for prototype development.

2 Marks

Sources/Platforms of IoT Development Boards

The following development boards are commonly used for IoT prototype development:

1. Arduino

- Easy to use microcontroller board
- Used for sensor interfacing and automation

2. Raspberry Pi

- Small single-board computer
- Supports Linux OS and IoT applications

3. NodeMCU (ESP8266)

- Wi-Fi enabled development board
- Used for wireless IoT projects

4. BeagleBone Black

- High-performance embedded board
- Suitable for industrial IoT

5. Intel Galileo / Edison

- Supports IoT and embedded computing

6. TI LaunchPad

- Low-power microcontroller platform by Texas Instruments

Q.1 (b)

(ii) List communication protocols used in IoT.

2 Marks

Communication Protocols Used in IoT

The commonly used communication protocols in IoT are:

1. **MQTT (Message Queuing Telemetry Transport)**
 - Lightweight publish/subscribe protocol
2. **CoAP (Constrained Application Protocol)**
 - Used for low-power devices
3. **HTTP/HTTPS**
 - Used for web communication
4. **Bluetooth Low Energy (BLE)**
 - Short-range wireless communication
5. **ZigBee**
 - Low-power mesh networking protocol
6. **Wi-Fi**
 - High-speed wireless communication
7. **LoRaWAN**
 - Long-range communication protocol
8. **6LoWPAN**
 - IPv6 over Low-Power Wireless Personal Area Network

Q.1 (c) Explain various levels of IoT.

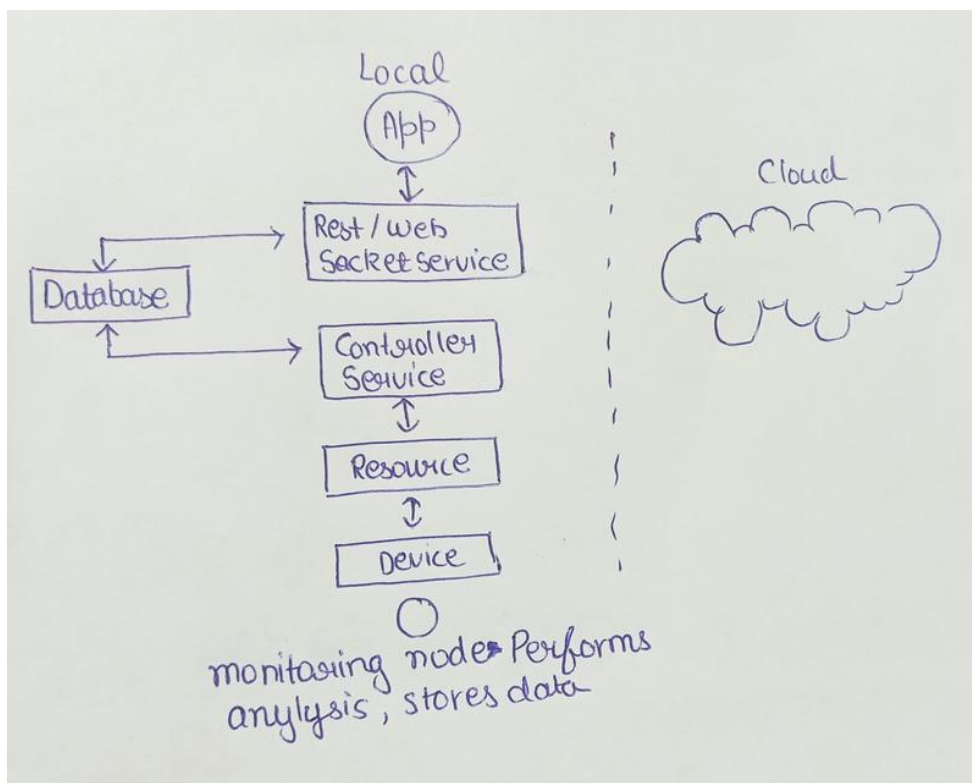
7 Marks

Various Levels of IoT

IoT systems are divided into different levels based on complexity, communication, and data processing capabilities.

Level 1

- In this level, the IoT system has a single node or device that performs sensing and actuation, stores data, performs analysis, and hosts the application.

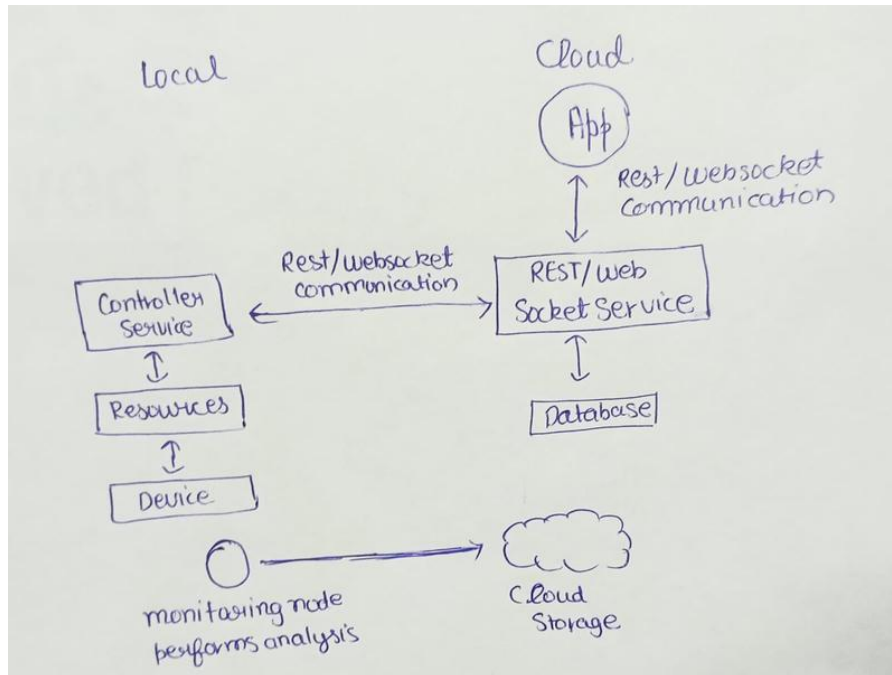


- The data sensed is stored locally.
- The data analysis is stored locally.
- Monitoring and control are done using a mobile app or a web app.
- The data generated in this level application is not huge.
- All the control actions are performed through the internet.
- Level 1 IoT systems are suitable for modelling low-cost and low complexity solutions where the data involved is not big and the analysis requirements are not computationally intensive.

Example: Room temperature is monitored using a temperature sensor, and data is stored and analysed locally.

Level 2

- In this level IoT system has single node that performs sensing / accuration and local analysis.
- Data is stored in the cloud, and the application is usually cloud-based.
- These are suitable for where the data involved is big; However, the primary analysis requirement is not computationally intensive and can be done locally.

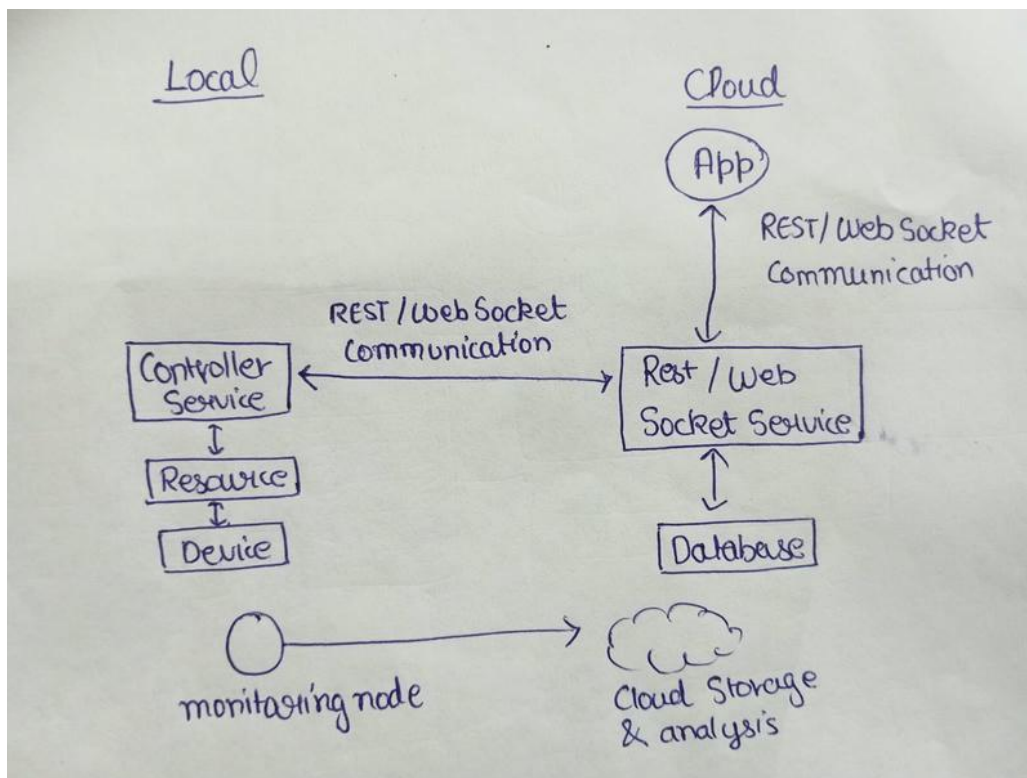


- This level consists of an air conditioner, temperature sensor, big data (bigger than level 1, data analysis done here) cloud, and control & monitoring app.
- This level 2 is complex compared to level 1. Moreover rate of sensing is faster than level 1.
- This level has a voluminous size of data. Hence, cloud storage is used.
- Data analysis is carried out locally. Cloud is used only for storage purposes.
- Based on data analysis, control action is triggered using a web app or a mobile app.

Example. Culture application, room freshening solution based on smell sensor, etc.

Level 3

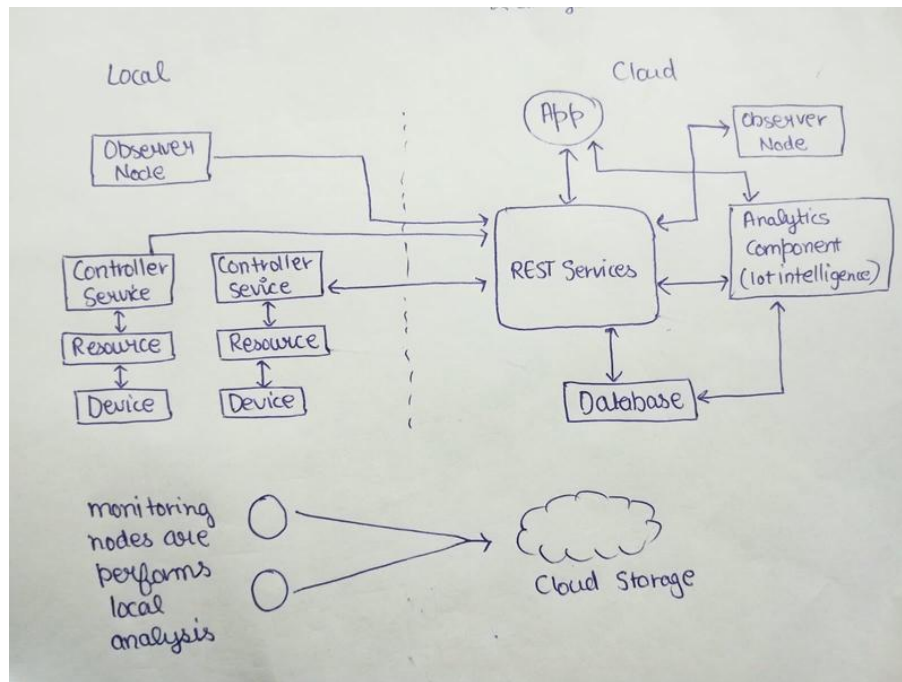
- In this level, the IoT system has a single node. Data is stored and analysed in the cloud, and the application is cloud-based.
- These level systems are suitable for solutions where the data involved is big and the analysis requirements are computationally intensive.
- Data here is voluminous i.e. Big data. Samsung is fast and collects sensed data on the cloud, because of which the amount of data increases rapidly.
- Data analysis is done on cloud size, and based on the analysis, control action is triggered using a mobile app or a web app.



Example. Tracking package handling, used accelerometer and gyroscope.

Level 4

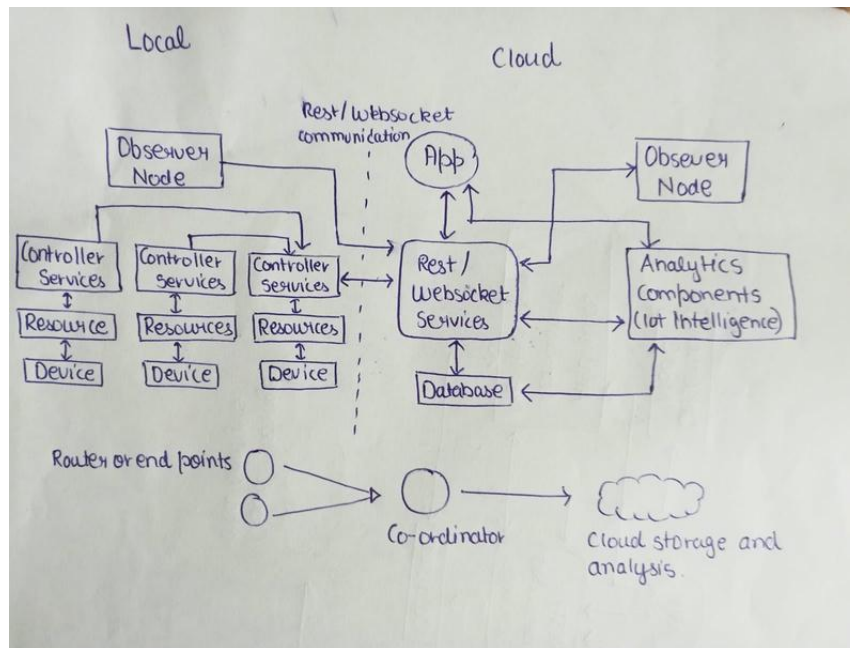
- In this level, the IoT system has multiple nodes that perform local analysis.
- Data is stored in the cloud, and the application is cloud-based.
- It contains local and cloud-based observer nodes that can subscribe to and receive information collected in the cloud from IoT devices.
- These types of IoT systems are suitable for solutions where the number of nodes is considerably high, the data involved is big, and the analysis requirements are computationally intensive.



Example. Noise monitoring, sound sensors are used.

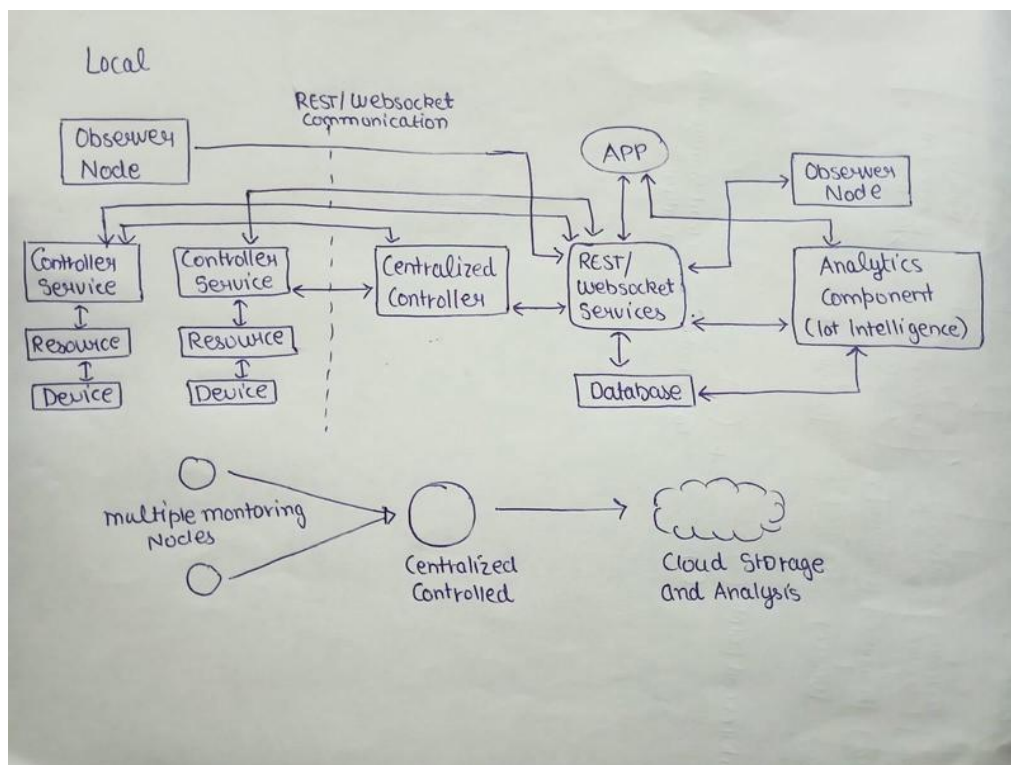
Level 5

- In this level, the IoT system has multiple nodes and one coordinator node.
- The endnodes perform sensing and actuation.
- The coordinator node collects data from the end node and sends it to the cloud.
- Data is stored and analysed in the cloud application, which is cloud-based.
- Level 5 iot systems are suitable for solutions based on wireless sensor networks, in which the data involved is used and the analysis requirements are computationally intensive.



Level 6

- At this level, the IoT system has multiple Independent and note that perform sensing or actuation and send data to the cloud.



- Data is stored in the cloud, and the application is cloud-based.
- The analytics component analyses the data and stores the results in a cloud database.
- The results are visualised with the cloud-based application.

- The centralised controller is aware of the status of all nodes and sends control commands to the nodes.

Advantages of IoT Levels

- Better scalability
- Faster communication
- Improved automation
- Efficient data management

Conclusion

Different IoT levels help in designing systems from simple smart devices to large-scale intelligent applications like smart cities and industries.